

Lab 5

Arithmetic Instructions

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Compare and Exchange Instruction in 80486 Microprocessor

CMPXCHG D, S - The instruction above will be evaluated like:

```
If A == D
    D = S
Else
    A = D
```

Note: Don't forget to change the processor while trying the instruction.

```
.686p
.model flat,stdcall
.stack 4096
ExitProcess proto,dwExitCode:dword

.data

.code
main proc

    MOV AL,5
    MOV CL,5
    MOV DL,1
    CMPXCHG CL,DL
    CMPXCHG CL,DL

    invoke ExitProcess, 0

main endp

end main
```

Arithmetic Instruction in 8086 Microprocessor

Following is the table showing the list of some more arithmetic instructions:

OPCODE	OPERAND	EXPLANATION	EXAMPLE
DAA	none	decimal adjust accumulator	DAA
DAS	none	decimal adjust accumulator after subtraction	DAS
AAA	none	ASCII adjust accumulator after addition	AAA
AAS	none	ASCII adjust accumulator after subtraction	AAS
AAM	none	ASCII adjust accumulator after multiplication	AAM
AAD	none	ASCII adjust accumulator after division	AAD

Here D stands for destination and S stands for source.
D and S can either be register, data or memory address.

1. DAA Example:

```
MOV AX, 38h
```

```
MOV BX, 45h
```

;You want to treat numbers as decimals and perform addition

```
ADD AX, BX
```

;AX will have hex addition, which is 7Dh, but I want decimal addition

```
DAA
```

;Now AX is adjusted to decimal addition which is 83

```

38 ---> 0011 1000
+ 45 ---> 0100 0101
-----
83    0111 1101
-----
      7  D

```

2. AAA Example:

ASCII Value of '5' is 35

ASCII Value of '2' is 32

I want to perform 5+2, but in my registers ASCII values are stored, since I took input from keyboard.

```
MOV AX, 35h
```

```
MOV BX, 32h
```

;Assume AX and BX values, you have got from keyboard and are stored in ASCII form

```
ADD AX, BX
```

;Now your AX is 37h, which is ASCII of '7'

```
AAA
```

;Now your AX has 7, it treats AX as ASCII number and converts back to character

3. AAM Example:

You want to perform $8 \times 4 = 32$ and print it.

```
MOV AX, 8
```

```
MOV BX, 4
```

```
MUL BL
```

;Now your AX is 20h, but you want ASCII value to be stored in AX after multiplication so that you can print that

```
AAM
```

;Now your AX has 0302, which is your result 32

Lab Work

1. Implement sample program given for CMPXCHG and explain how the instruction is working. Attach screenshot of register window after execution of each CMPXCHG instruction.
2. Implement the program that demonstrates the following instructions, given AX and BX register values.

Take reference from sample program given in the manual.

Attach the code with comments on each adjust instruction, and attach screenshot of the output (AX register mostly) after each adjust instruction.

Write how operation 5 and 7 (given in table below) are performed in detail. (i.e., how value in AX is interpreted after the operation)

Note: For AAD, first you have to perform AAD, then division operation

	OPCODE	AL	BL	OPERATION
1	DAA	05h	05h	ADD AL, BL
2	DAS	20h	05h	SUB AL, BL
3	AAA	35h	32h	ADD AL, BL
4	AAS	38h	32h	SUB AL, BL
5	AAS	32h	38h	SUB AL, BL
6	AAM	11d	03d	MUL BL
7	AAD	AX = 0408h	04h	AAD; DIV BL; DAA